Resolution in Propositional Logic

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What is a Literal?

A literal is:

- A propositional variable (P)
- Its negation ($\neg P$)

Examples:

$$P, \quad \neg Q, \quad R$$

What is a Clause?

A clause is a disjunction (OR) of literals.
Examples:

$$P \vee Q$$

$$\neg P \vee R$$

$$P \vee Q \vee R$$

Resolution works only on clauses.

Resolution Rule

If we have:

$$P \vee Q$$

$$\neg P \vee R$$

We resolve on P and derive:

$$Q \vee R$$

Explanation: Complementary literals cancel.

Basic Inference Rules (Review)

Modus Ponens

$$p \to q, \quad p \Rightarrow q$$

Modus Tollens

$$p \to q, \quad \neg q \Rightarrow \neg p$$

Hypothetical Syllogism

$$p \to q, \quad q \to r \Rightarrow p \to r$$

Disjunctive Syllogism

$$p \vee q, \quad \neg p \Rightarrow q$$

Practice 1 (With Solution)

Given:

$$P \vee Q$$

$$\neg P$$

Resolution:

$$Q$$

Explanation: $\neg P$ eliminates P, leaving Q.

Practice 2 (With Solution)

Given:

$$P \vee Q$$

$$\neg Q \vee R$$

Resolution on Q:

$$P \vee R$$

Explanation: Q and $\neg Q$ cancel.

Resolution Method

Resolution is:

- An inference rule
- Sound and complete for propositional logic
- Used in automated theorem proving

To prove a goal, we use **Proof by Refutation**.

Proof by Refutation Strategy

To prove α:

1. Convert Knowledge Base to CNF
2. Add $\neg\alpha$ (Goal)
3. Apply resolution
4. Derive empty clause $\perp$

If $\perp$ is derived $\rightarrow$ original goal is true.

Conjunctive Normal Form (CNF)

CNF = AND of ORs.

Steps:

1. Eliminate $\leftrightarrow$
2. Eliminate $\rightarrow$
3. Push negation inward (De Morgan)
4. Apply distributive law

Disjunctive Normal Form (DNF)

Definition: DNF = OR of ANDs

$$(p \wedge q) \vee (\neg p \wedge r)$$

Each bracket is a conjunction (AND of literals). The entire expression is a disjunction (OR of terms).

Example of conversion:

$$p \wedge (q \vee r)$$

$$\Rightarrow (p \wedge q) \vee (p \wedge r)$$

Now it is in DNF.

DNF vs CNF

DNF = OR of ANDs.
Resolution works on CNF only.

Complete CNF Conversion Procedure

To apply Resolution, we must first convert formulas to **Conjunctive Normal Form (CNF)**.

Step 1: Eliminate Biconditionals

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Step 2: Eliminate Implications

$$p \rightarrow q \equiv \neg p \vee q$$

Step 3: Push Negation Inward (Using De Morgan's Laws)
Step 4: Apply Distributive Law Convert into AND of OR clauses.

Final form: Conjunction of disjunctions of literals.

Problem 1

Premises:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. P

Prove: R

Solution: Convert to CNF

1. $\neg P \vee Q$
2. $\neg Q \vee R$
3. P

Add negation of goal:

4. $\neg R$

Resolution Steps

Resolve (2) and (4):

$$\neg Q$$

Resolve (1) and (3):

$$Q$$

Resolve Q and $\neg Q$:

$$\perp$$

Since contradiction obtained, R is true.

Problem 2

1. Humidity high OR sky cloudy 2. If sky cloudy $\rightarrow$ rain 3. If humidity high $\rightarrow$ hot 4. Not hot Goal: Rain

Symbolic Representation

Let: P: Humidity high Q: Sky cloudy R: Rain S: Hot

Before Simplification (Implication Form):

$$P \vee Q$$
$$Q \rightarrow R$$
$$P \rightarrow S$$
$$\neg S$$

After Removing Implications (CNF Form):

$$P \vee Q$$
$$\neg Q \vee R$$
$$\neg P \vee S$$
$$\neg S$$

Resolution Steps

Resolve $\neg Q \vee R$ and $\neg R$:

$$\neg Q$$

Resolve $P \vee Q$ and $\neg Q$:

$$P$$

Resolve $\neg P \vee S$ and P:

$$S$$

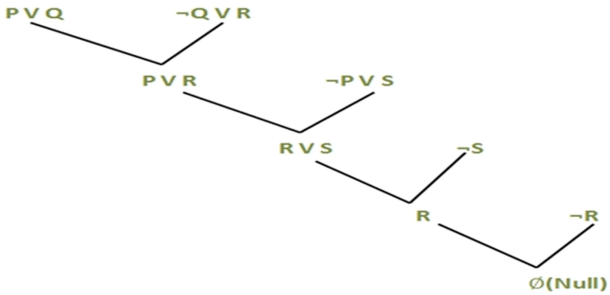
Resolve S and $\neg S$:

$$\perp$$

Therefore, Rain is true.

Resolution Graph

Finally, apply the rule as shown below:



After applying Proof by Refutation (Contradiction) on the goal, the problem is solved, and it has terminated with a **Null clause (∅).** Hence, the goal is achieved. Thus, It is not raining.

By Mukesh Barapatre

Problem 3

Given the following hypotheses:

1. If it rains, Joe brings his umbrella $(r \rightarrow u)$
2. If Joe has an umbrella, he doesn't get wet $(u \rightarrow \neg w)$
3. If it doesn't rain, Joe doesn't get wet $(\neg r \rightarrow \neg w)$

Prove that Joe doesn't get wet: $(\neg w)$

Convert to CNF

Before Simplification (Implication Form):

$$r \rightarrow u$$

$$u \rightarrow \neg w$$

$$\neg r \rightarrow \neg w$$

After Removing Implications (CNF Form):

$$\neg r \vee u$$

$$\neg u \vee \neg w$$

$$r \vee \neg w$$

(All implications removed)

Resolution by Refutation (Systematic Method)

Goal: Prove $\neg w$
Add negation of goal:

$$w$$

Clause set now:

$$\neg r \lor u$$

$$\neg u \lor \neg w$$

$$r \lor \neg w$$

$$w$$

Refutation Resolution Steps

From (2) and w:

$$\neg u$$

From (1) and $\neg u$:

$$\neg r$$

From (3) and $\neg r$:

$$\neg w$$

From $\neg w$ and w:

$$\perp$$

Empty clause derived.
Therefore $\neg w$ is true.

Resolution in AI

- Used in propositional and predicate logic
- Used for satisfiability testing
- Also called Proof by Refutation
- Null clause indicates contradiction
- Resolution is sound and complete